

Chiller Operations & Maintenance Guide

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Overview

This guide provides operational procedures and maintenance schedules for centrifugal and screw chillers used in HVAC systems. Follow all guidelines to ensure optimal performance, safety, and equipment longevity.

1. Operating Parameters (Section 7)

1.1 Chilled Water Temperature

Target leaving water temperature: **6-10°C (43-50°F)**

Supply water setpoint: $7^{\circ}\text{C} \pm 1^{\circ}\text{C}$ ($44^{\circ}\text{F} \pm 2^{\circ}\text{F}$)

Return water expected: $12-18^{\circ}\text{C}$ ($54-64^{\circ}\text{F}$)

Temperature differential: Minimum 4°C (7°F) for efficient operation

Emergency shutdown: Below 2°C (ice formation risk) or above 25°C (pressure relief)

1.2 Condenser Water Temperature

Entering condenser temperature: **25-32°C (77-90°F)**

Leaving condenser: $3-5^{\circ}\text{C}$ above entering temperature

Approach temperature: $5-8^{\circ}\text{C}$ above ambient wet bulb

If temperature too high: Increase cooling tower fan speed or flow rate

1.3 System Pressure

Evaporator pressure: 200-350 kPa (29-51 psi) depending on refrigerant type

Condenser pressure: 800-1200 kPa (116-174 psi)

Pressure differential: Maintain stable differential for efficient cycle

If pressure drops: Check for refrigerant leaks immediately

1.4 Flow Rates

Chilled water flow: 0.05-0.10 m³/s per ton of refrigeration

Condenser water flow: 0.08-0.15 m³/s per ton

Minimum flow: 0.03 m³/s (to prevent freeze-up in winter)

Flow measurement: Check differentials across heat exchangers <2 psi when clean

2. Startup Procedure

1. Verify oil level in compressor crankcase (sight glass at 50% mark)
2. Check refrigerant charge (sight glass should show 10-15% subcooling)
3. Enable cooling tower fan and verify water circulation

4. Set chilled water setpoint to 10°C (gradual pull-down recommended)
5. Energize chiller control and slowly throttle chilled water valve
6. Monitor pressures for stability (typically stabilizes in 10-15 minutes)
7. Verify all safety interlocks responding correctly
8. Record starting amperage on all compressor motor phases

3. Maintenance Schedule (Section 9)

Maintenance Task	Frequency	Priority
Filter inspection	Weekly	High
Water treatment check	Weekly	High
Oil level check	Monthly	Medium
Pressure & temperature log	Daily	High
Vibration monitoring	Monthly	Medium
Heat exchanger cleaning	Annually	High
Refrigerant system test	Every 2 years	Medium
Bearing lubrication	Every 6 months	Medium

4. Common Issues & Resolution

Issue: Chiller not reaching setpoint temperature

Causes: Low refrigerant charge, fouled heat exchangers, compressor valve leak

Resolution: Check superheat, clean condenser tubes, call service if refrigerant low

Issue: High discharge pressure (>1200 kPa)

Causes: Condenser fouling, high ambient temperature, non-condensable gases

Resolution: Clean condenser, increase cooling tower flow, purge system

Issue: Oil migration from compressor

Causes: Low evaporator pressure, improper oil separator function, viscosity mismatch

Resolution: Check oil separator drain, verify evaporator superheat $>4^{\circ}\text{C}$